Dynamics of networks – Mini Project Assignment

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You have received by email one data table (in standard text delimited format ".csv") which provides you yearly information on dyads (couples) of individual agents and the groups they are involved in. Each agent of the dyad may be part of several groups, and then the dyad is duplicated (in line) for each group of each partner. One variable (link) indicates if the partners in the dyad collaborate that year. Only the dyads involving individuals who have at least once collaborated (link is at least once equal to 1) are listed.

Objectives

Some groups have applied to be funded by a public policy, and part of them have been selected in year 2011, and benefited from the policy since then. This policy involved money transfers every year after year 2012 (included). The goals of the policy are to improve individual outcomes and stimulate collaborations.

In your essay and your presentation documents, you will:

1. Describe your dataset;
2. Stress hypotheses using relevant theoretical models (see chapters of the course);
3. Use relevant econometric techniques (see below) to answer the assignments that have been given to you.

Assignments: you will treat the two assignments

a) **Assignment 1:**
   1. You will assess (estimate) the average impact of the policy on the yearly individual outcome variables of the individuals who are members of treated groups.
   2. The impact will be observed on the yearly individual outcomes variables, the number of links internal to the group, the number of links outside the group, and the 3 outcome variables.
   3. You will also assess (estimate) the average impact of the policy of the individuals who are NOT members of treated groups but have been collaborators of members of treated groups before treatment year.
   4. You will compare and discuss those effects, in relation with the course.
   5. You will answer to the same questions by distinguishing females and males.
   6. You will answer to the same questions by distinguishing “core” and “non-core” members of groups.

b) **Assignment 2**
   1. You will assess (estimate) the average impact of the policy (one or the two being treated) on the likelihood (linear probability approach – as in a linear regressions) that agents form connections.
   2. You will assess (estimate) the average impact of the policy on the likelihood (linear probability approach) that agents form connections, when they are within the same group, or outside the same group.
   3. You will answer to the same questions by distinguishing “core” and “non-core” members of groups (3 possibilities: “core” and “non-core; “core” and “core”; “non-core” and “non-core”).
   4. You will answer to the same questions by distinguishing females and males.

Method for impact analysis

You are supposed to conduct an impact analysis consists, essentially in comparing similar "treated" and "relevant non-treated" (controls) persons after the treatment year. This is the diff-in-diff approach. Here treatment intervened in 2012, so that:

- Before treatment ⇔ year<2012;
- After treatment ⇔ year>=2012.

The selection among groups may be considered nearly as random (as an experiment) conditional on selection criterion =1 (see variables below, you have two criteria available you may play with). This is essential for the diff-in-diff approach to be consistent. The validity of this statement may however be challenged (and discussed) by comparing treated and controls before treatment. If selection is nearly as random conditional on selection, then treated and controls (non-treated such that selection =1) likely "look the same" and have similar trends before treatment. So you may exclude observations that do not meet the selection criterion (you will have to decide how to work with this depending on your assignment).

1/ Typically, if you deal with the impact of the treatment on individual outcomes (assignments a and b), you need to transform your dataset to work with a data table having as an observation (a line) an individual *i* in a given group *g* observed a given year *t*. Then you need to perform regressions of the form:

$$y_{i,g,t} = \alpha + \beta T_{g_i} \times Post_t + \gamma X_{i,t} + \delta_t + \theta_i + \varepsilon_{i,g,t},$$

where $y_{i,g_i,t}$ is an outcome variable, T_g is the treatment variable (1 if treated and 0 otherwise, and is specific to the group if i, g), $Post_t$ is the post treatment period (to be created, cf. above), $X_{i,t}$ is a vector of control variables, θ_i is the individual fixed effect (a dummy per individual i), δ_t is a year fixed effect (a dummy per year t), and $\varepsilon_{i,g,t}$ is the error term.

You may take into account that $\varepsilon_{i,g,t}$ and $\varepsilon_{i,g',t}$ are likely correlated (typically referred to as "cluster" option in econometric softwares). This is important for inference only (statistical significance) as does not affect coefficients estimates).

You are interested in β estimates that will give you the impact of treatment. Interpretation should be made comparing $\hat{\beta}$ (impact in absolute terms) with the average y.

Warning: You have a dataset of dyads that needs to be transformed before running regressions. Note those dyads are bilateral (when dyad *ij* appears, then dyad *ji* does not). So performing regressions at the individual level requires first to switch identities in an instance of the dataset and then append it to the dataset itself, and second you need to reduce the dataset, from 1 line = a dyad of group*individuals*year to 1 line = a group*individuals*year.

2/ If you are dealing with the impact of the treatment on link formation (assignment c), you can use directly the transmitted dataset, and perform a variant of the above regression, of the form (called linear probability approach on dyads):

$$y_{ij,g_i,g_j,t} = \alpha + \beta \left|T_{g_i} - T_{g_j}\right| \times Post_t + \lambda(T_{g_i} \times T_{g_j}) \times Post_t + \gamma X_{ij,t} + \delta_t + \theta_{ij} + \varepsilon_{ij,g_i,g_j,t},$$

with $y_{ij,g_i,g_j,t}$ the dyad link dummy, $X_{ij,t}$ is a vector of (dyad specific) control variables, θ_{ij} is the dyad fixed effect (a dummy per dyad *ij*), δ_t is a year fixed effect (a dummy per year t), and $\varepsilon_{ij,g_i,g_j,t}$ is the error term.

You are here interested in β estimates that will give you the impact of (only) one agent being treated, and in λ estimates that will give you the impact of both agents being treated, on the frequency of being connected. Interpretation should be made comparing $\hat{\beta}$ and $\hat{\lambda}$ (impacts in absolute terms) with the average y.

You may take into account that $\varepsilon_{ij,g_i,g_j,t}$ and $\varepsilon_{ij',g_i',g_{j'},t}$ are likely correlated, that is for the same one agent in different groups, and for the same agent with different partners. And this for both sides of the dyad (two-way fix effect, and with option cluster). Again, this is important for inference only (statistical significance) as does not affect coefficients estimates).

One problem with this regression is that if $\left|T_{g_i} - T_{g_j}\right| = 1$ then $T_{g_i} \times T_{g_j} = 0$, and vice-versa (two, one or zero treatments on the dyad are mutually exclusive). Then, the third modality if that none of the two persons, *i* and *j* have been treated which will be taken into reference.

Tools

All pictures and networks analyses **can be performed in R or Pyhton**.

For the impact analysis, you should use **regression packages that allow you to include many fixed effects.** You should rely on specific commands that are available on the following statistical softwares:

- In Stata, you may use **reghdfe** package (install code: "ssc install reghdfe").
- In R, you may use **fixest** (see https://lrberge.github.io/fixest/).

Variables in the dataset

- id_dyad_in_group: unique identifier of the dyad in a given pair of groups (one for each partner
- year: the calendar year
- Id_1 : individual identifier of first member
- Id_2 : individual identifier of second member
- yearly_collabs_dyad: counts the number of collaborations between the two members of the dyad that year
- group_id_1 : group identifier of first member; when missing, this means that the first agent was not involved in any group that applied to the policy
- group_id_2 : group identifier of second member; when missing, this means that the second agent was not involved in any group that applied to the policy
- group_treated_1 : =1 if the group of member 1 was treated by the policy, =0 if the group applied but was not selected into treatment; when missing, this means that the first agent was not involved in any group that applied to the policy.
- group_treated_2 : =1 if the group of member 1 was treated by the policy, =0 if the group applied but was not selected into treatment; when missing, this means that the second agent was not involved in any group that applied to the policy.
- outcome_X_1 : = individual outcome X (=1,2,3) of first member
- outcome_X_2 : = individual outcome X (=1,2,3) of second member
- female_1 : = 1 if member 1 is female female, = 0 if member 1 is male
- female_2 : = 1 if member 2 is female female, = 0 if member 2 is male
- core_1 : dummy indicating that member 1 is an important player in the group goup_id_1 with respect to the policy
- core_2 : dummy indicating that member 2 is an important player in the group goup_id_1 with respect to the policy
- treated_1: =1 if the group of member 1 has been treated
- treated_2: =1 if the group of member 2 has been treated
- age_1 : = age of the first member
- age_2 : = age of the second member
- selection1_1 := 1 means agent 1 respects the main selection criterion that is required for comparing outcomes, on the group of member 1; when missing, this means that the first agent was not involved in any group that applied to the policy. It is NOT the jury selection, just a variable something you will use to compare treated and non treated.
- selection1_1 := 1 means agent 2 respects the main selection criterion that is required for comparing outcomes, on the group of member 2; when missing, this means that the first agent was not involved in any group that applied to the policy. It is NOT the jury selection, just a variable something you will use to compare treated and non treated.
- selection2_1 := respects an alternative selection criterion that is required for comparing outcomes, on the group of member 1; when missing, this means that the second agent was not involved in any group that applied to the policy.
- selection2_2 := respects an alternative selection criterion that is required for comparing outcomes, on the group of member 2; when missing, this means that the second agent was not involved in any group that applied to the policy.

- Field_X_1: Field X dummy of agent 1
- Field_X_1: Field X dummy of agent 2

Some variables you may have to create (among others):

- id_dyad: unique identifier of the dyad
- link_dyad: =1 of yearly_collabs_dyad>0
- agemax :=max(age_1,age_2)
- agemean :=mean(age_1,age_2)
- agedif :=abs(age_1-age_2)
- Post := 1 if year >= 2012

Grading Criteria

- ability to stress clearly expressed and specific questions (a danger is to be only descriptive),
- ability to connect your questions with network theories presented in the course,
- ability to use relevant empirical methods,
- ability to make statements based on the data and your empirical investigations, and
- ability to discuss the limitations of your results,
- ability to provide a neat and clear writing (in English),
- quality of the code (and proper comments inside) that one should be able to run without bugs.
- quality of the presentation documents (slides and accompanying text): good starting statement, good selection of ideas to focus on, neat presentation slides and bold conclusion!

You should provide

- **A 20 pages long report** depending on the size of the fonts and spacing between lines,
- **Slides of the oral presentation and an accompanying text** as a support of your presentation (in bullet list form),
- **The code (and proper comments inside) that reproduces your results** and that anyone should be able to run without bugs once appropriate paths to the data and to the outcome folders are specified.